

PROFILE SUMMARY

- Lead Automation Test Engineer with 9+ years of comprehensive experience spanning industrial automation, PLC–SCADA systems, electrical and control engineering, and enterprise-scale software testing across automotive, wind energy, biotech, pharmaceutical, and heavy manufacturing domains.
- I am currently working in an IT-centric automation testing role, responsible for end-to-end validation of SCADA applications, Beckmann PLC logic, and Hardware-in-the-Loop (HIL) systems for real-time wind turbine monitoring and control platforms.
- Possesses deep expertise in testing and validating real-time control systems, including PLC I/O verification, control logic testing, alarm and event validation, fail-safe and interlock testing, and safety-critical control scenarios.
- Strong hands-on experience in SCADA functional testing, dashboard validation, telemetry accuracy verification, and secure control command execution under manual and semi-automatic operating modes. Well-versed in OT–IT integration, ensuring reliable communication between PLCs, SCADA layers, backend services, and web-based user interfaces.
- Highly experienced in HIL testing methodologies, simulating real-world operational conditions to validate system behavior, performance, and reliability before deployment. Skilled in validating high-frequency telemetry pipelines using industrial and IT protocols such as MQTT, OPC UA, IEC-61850, and WebSockets, ensuring data integrity, low latency, and system stability in distributed environments.
- Strong background in software and backend validation, working closely with development teams to test Python and Flask-based microservices, REST APIs, and real-time data processing layers backed by MongoDB and Cassandra. Adept at analyzing logs, identifying root causes, reproducing defects, and supporting rapid issue resolution across development, testing, and production environments.
- Brings extensive prior experience in automation project installation, commissioning, FAT/SAT, instrument calibration, and production support, enabling a holistic understanding of the full system lifecycle from field devices to enterprise applications.
- Proven ability to lead technical teams, coordinate cross-functional stakeholders, prepare detailed test documentation, and support global deployments. Recognized for a strong problem-solving mindset, ownership attitude, and ability to bridge the gap between industrial automation and modern IT systems.

TECHNICAL SKILLS

- **Automation & Control:** PLC–SCADA Systems, Beckmann PLC, OPC UA, IEC-61850, HMI, Industrial Networking, Servo Systems
- **Testing & Validation:** Automation Testing, PLC Testing, HIL Testing, System Integration Testing, Functional Testing, Instrument Validation, Commissioning Testing
- **Software & IT:** Python, Flask, React.js, JavaScript, REST APIs, WebSockets, MQTT
- **Databases;** MongoDB, Cassandra
- **Industrial Domains;** Wind Energy, Automotive Manufacturing, Biotech & Pharmaceuticals, Robotics & Conveyors
- **Professional Skills;** Troubleshooting, Root Cause Analysis, Team Leadership, Documentation, Customer Coordination
- **Others:** Jira, Agile Methodologies, GitHub, Postman,

PROFESSIONAL EXPERIENCE

Vividnex Innovation Pvt. Ltd.
Lead Test Engineer – Automation & SCADA

Nagpur MH
01 July 2023 – Present

Roles & Responsibilities:

- Leading automation testing and validation activities for PLC–SCADA based industrial software platforms.
- Responsible for Beckmann PLC testing, including:
 - Digital & analog I/O validation
 - Control logic verification
 - Alarm, trip, and interlock testing
 - Emergency stop and fail-safe condition testing
- Executing Hardware-in-the-Loop (HIL) testing by simulating turbine and grid behavior to validate PLC and SCADA response.
- Performing SCADA functional testing, including:
 - Dashboards and UI validation
 - Alarm & event management
 - Trend and historical data verification
 - Control command execution and acknowledgment
- Validating manual, semi-automatic, and remote-control modes with safety interlocks, dead-man timers, and RBAC.
- Testing real-time telemetry pipelines using MQTT, OPC UA, WebSockets, and IEC-61850 protocols.
- Supporting backend and API testing for Python & Flask services handling telemetry ingestion and processing.
- Verifying data integrity and performance in MongoDB and Cassandra for high-frequency industrial data.
- Performing system integration testing across PLC, SCADA, backend services, and web applications.
- Preparing test cases, test plans, validation reports, defect logs, and release documentation.
- Coordinating with developers, automation engineers, and DevOps teams to resolve defects.

Roles & Responsibilities:

- Managed onsite installation, commissioning, and validation of industrial automation systems for biotech and pharmaceutical plants.
- Prepared BOMs for control panels, cables, glands, lights, sockets, and earthing systems.
- Designed and reviewed PLC-based control systems, hardware selection, I/O devices, and communication protocols.
- Integrated PLC systems with HMI, SCADA, sensors, actuators, and safety systems.
- Performed system testing, loop checks, and functional validation before production handover.
- Conducted precision instrument assembly, calibration, inspection, validation, and maintenance.
- Executed FAT, SAT, and commissioning activities as per GMP standards.
- Handled breakdown maintenance, preventive maintenance, and troubleshooting.
- Led daily team meetings, work allocation, and critical re-test discussions.
- Coordinated with cross-functional teams to improve production efficiency through automation.
- Provided production support and system documentation.
- Supported automation system troubleshooting, breakdown analysis, and preventive maintenance.
- Coordinated daily site activities, led technical teams, and participated in critical re-test and validation meetings.
- Provided production support and documentation for automation systems.

Automag India Private Limited

Senior Automation Engineer

Pune, MH

09 September 2019 – 09 April 2021

Roles & Responsibilities:

- Installed and commissioned automated assembly lines, conveyors, press machines, and robotic cells.
- Installed and commissioned automated assembly lines, conveyors, press machines, and robotic cells.
- Specialized in material handling automation and precision assembly systems.
- Performed electrical installation, wiring checks, signal testing, and system validation.
- Supported PLC I/O verification and basic logic testing during commissioning.
- Executed new machine bring-up, testing, and commissioning.
- Handled AMC, breakdown maintenance, and preventive maintenance.
- Coordinated shutdown activities at various OEM customer locations.
- Supported troubleshooting and corrective actions during production issues.

Wipro PARI

Member – Field Services (Automation & Robotics)

Pune, MH

05 February 2018 – 09 September 2019

Roles & Responsibilities:

- Installed and commissioned automated press machines for engine and gearbox assembly lines.
 - Specialized in robotic cell installations, including Material handling, Spot & arch welding, Sunroof and wheel-arch hemming.
 - Installed and tested roller, slat, EMS, and overhead conveyor systems.
 - Conducted dry runs, functional testing, and production trials.
 - Provided AMC, breakdown maintenance, and client support.
- Led site teams and coordinated daily installatio activities.

KEY PROJECT & CUSTOMER EXPERIENCE

- Bajaj Chetak EV Plant, Pune – Overhead conveyor AMC & maintenance
- Bharat Biotech (Covaxin Project) – Automation installation & commissioning
- Suzuki Motors Gujarat – White body slat conveyor system
- Renault-Nissan Chennai – Engine & gearbox assembly lines, servo press systems
- Ford India (Chennai & Sanand) – Robotic welding & hemming cells
- Daimler India (BharatBenz) – Vehicle & powertrain assembly lines
- Komatsu India, Royal Enfield, Yamaha, Swaraj Engines, Escorts-Kubota – Automation installations & AMC

PROJECTS

SCADA Online Control & Monitoring System for Wind Turbines

Company: Vividnex Innovation Pvt. Ltd.

(Bangalore)Client Location

July 2023 – Present

Technology Stack: Python, Flask, React.js, JavaScript, REST APIs, WebSockets, MQTT, PLC–SCADA, OPC UA, IEC-61850, MongoDB, Cassandra, Beckmann PLC, HIL

Project Description:

Developed and tested an enterprise-grade, web-based SCADA platform designed for real-time monitoring, control, diagnostics, and analytics of wind turbines deployed across multiple wind farms. The system captures high-frequency telemetry data such as wind speed, rotor speed, power generation, temperature, vibration, and fault signals from PLCs and field devices. It provides centralized dashboards for turbine health monitoring, alarm management, performance analytics, power curves, wind distribution, and production reports. The platform supports secure remote operations with role-based access control, safety interlocks, and controlled manual/semi-

automatic operating modes to prevent unsafe actions. Designed for high availability and scalability to support global wind turbine operations.

Role & Working:

- Performed Beckmann PLC testing, including control logic validation, I/O signal verification, alarm handling, and fail-safe behavior testing.
- Executed Hardware-in-the-Loop (HIL) testing by simulating real turbine operating conditions and fault scenarios.
- Validated SCADA dashboards, real-time trends, alarm workflows, and historical reports for accuracy and reliability.
- Tested real-time telemetry pipelines using MQTT, OPC UA, IEC-61850, and WebSockets.
- Verified secure control commands, safety interlocks, dead-man timers, and role-based permissions.
- Supported backend and API testing for Python/Flask services and validated data storage in MongoDB and Cassandra.
- Prepared detailed test cases, validation documents, defect reports, and release readiness reports.

Covaxin Manufacturing Automation System
Company: Bharat Biotech (BioVet Unit – Malur, Karnataka)

May 2021 – June 2023
Domain: Pharmaceutical | Biotech

Technology Stack: PLC, HMI, SCADA, Industrial Instruments, Control Panels, Sensors & Actuators
Project Description:

Automation and control system implemented for Covaxin vaccine manufacturing, supporting critical pharmaceutical production processes under strict regulatory and GMP compliance. The project involved automation of process equipment, monitoring systems, and instrumentation used in vaccine production, ensuring high accuracy, traceability, and operational safety. The system enabled real-time monitoring of process parameters, alarms, and interlocks to maintain product quality and regulatory adherence. Extensive validation and documentation were required to meet pharmaceutical compliance standards.

Role & Working:

- Installed and commissioned PLC-based automation systems for vaccine manufacturing processes.
- Performed system testing, loop checking, and functional validation of control logic and field instruments.
- Executed instrument calibration, inspection, and validation activities as per GMP guidelines.
- Supported FAT, SAT, and commissioning with quality and production teams.
- Identified and resolved automation and instrumentation issues during production trials.
- Prepared validation records and supported production handover and audits.

Automotive Assembly Line Automation (Multiple OEMs)
Company: Renault-Nissan Automotive India Pvt. Ltd., Ford India, Daimler India, Suzuki Motors, Gujarat

February 2018 – April 2021
Domain: Automotive Manufacturing | Robotics

Technology Stack: PLC Systems, Robotics Cells, Servo Presses, Conveyor Systems (Roller, Slat, EMS, Overhead), SCADA
Project Description:

Automation of engine, gearbox, and vehicle assembly lines for leading automotive OEMs. The project involved installation and commissioning of robotic welding cells, material handling systems, servo press stations, and conveyor-based assembly lines. Systems were designed to ensure high throughput, precision assembly, safety compliance, and minimal downtime. Automation systems were integrated with PLCs and SCADA for monitoring, diagnostics, and production control.

Role & Working:

- Installed and commissioned robotic cells for spot welding, arch welding, hemming, and material handling.
- Performed functional testing, dry runs, and production trials of automated assembly systems.
- Supported PLC signal testing, I/O verification, and troubleshooting during commissioning.
- Conveyor systems installed and tested, including EMS and overhead conveyors.
- Handled shutdown activities, AMC, breakdown maintenance, and preventive maintenance.
- Coordinated with OEM teams to resolve automation issues during ramp-up and production phases.

EDUCATION & CERTIFICATIONS

- Diploma in Electronics Engineering (2013)
Government Polytechnic Barabanki – BTEUP
- Bachelor of Arts (2011)
Dr. Ram Manohar Lohia Avadh University, Ayodhya
- Diploma in Computer Applications (2008)
Datamation Foundation Charitable Trust, Barabanki
- Intermediate (12th) – 2007
Jawahar Navodaya Vidyalaya, Barabanki
- High School (10th) – 2005
Jawahar Navodaya Vidyalaya, Barabanki

DECLARATION

I hereby declare that the above information is true and correct to the best of my knowledge and belief.

Ramu Singh